

Task assignment as dynamic incentives^{*}

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Abstract

We study repeated task assignment as an instrument to provide effort incentives. Unlike traditional instruments, task assignment is inseparable from production and rivalrous: it both provides incentives and determines who produces, while necessarily excluding others. We show that workers are optimally assigned the task through a strict, evolving priority ranking. In every continuation, workers' expected average workloads differ, even when they are independent and symmetric in all aspects. Consequently, our results highlight that the efficiency-equality tension is more pervasive than previously recognized. We further examine design instruments that expand the scope of efficiency and the corresponding consequences on worker inequality.

Keywords: dynamic allocation, repeated games.

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1 Introduction

Organizations routinely assign workers to tasks: sales managers allocate territories, consulting firms staff projects, and gig-economy platforms dispatch drivers. Prior work studies task assignment largely through the lens of allocative efficiency, viewing it as a tool to leverage comparative advantage across heterogeneous workers; see, e.g., [Gibbons and Waldman \(1999\)](#) and [Garicano and Van Zandt \(2013\)](#) for surveys. This perspective abstracts from a key feature of many organizations: task assignment is a powerful incentive instrument, especially when monetary rewards are constrained. Desirable assignments such as flagship projects or lucrative routes reward strong performance, while undesirable ones discipline underperformance.

In this paper, we characterize optimal incentive provision through task assignment and show that it departs sharply from standard predictions of incentive theory. Even when workers are independent and symmetric in all respects, optimal incentives induce a strict ranking of their expected workloads in the future. Unequal workload may therefore reflect optimal incentives rather than worker heterogeneity ([Board, 2011](#); [Andrews and Barron, 2016](#)), and that the efficiency-equality tension is more pervasive than established ([Winter, 2004](#); [Halac, Lipnowski and Rappoport, 2021](#)). We further find that improving effort detection may perversely weaken effort incentives, unlike in canonical long-run relationships ([Kandori, 1992](#)). In contrast, improving shirking detection is superior for strengthening incentives through task assignment without exacerbating worker inequality, relative to other alternatives.

We study a canonical dynamic moral hazard setting without monetary transfers. A manager repeatedly interacts with a set of ex ante identical workers. We begin with a baseline model where the manager has a single task that is undesirable for the workers. In each period, she assigns one of the workers to perform this task. The assigned worker privately chooses whether to exert effort or to shirk, producing a public noisy output; efficiency requires that this worker exerts effort. Workers prefer to be unassigned. When assigned, workers strictly prefer to shirk than to exert effort; in contrast, the manager prefers the efficient outcome. The manager’s strategy, which we call an assignment rule, is flexible: in each period, her assignment can depend arbitrarily upon all past outputs and assignments.

We characterize all uniformly efficient assignment rules for any given number of workers; these rules sustain an efficient equilibrium simultaneously for all parameters—output noise, payoffs, and discounting—given which an efficient equilibrium exists. This uniformity criterion

is practically appealing: it precludes fine-tuned assignment in the spirit of the Wilson doctrine (1987), acknowledging that in practice, managers, whether human or algorithmic, rarely observe the precise details of agency relationships or continuously recalibrate incentives as those details evolve. Crucially, this criterion entails no loss for efficiency: by definition, a uniformly efficient rule sustains an efficient equilibrium whenever such equilibrium exists.

As an incentive instrument, task assignment differs from canonical ones such as monetary transfers or information design. First, it is inseparable from production: assigning the task to a worker requires simultaneously motivating his effort. Second, it is rivalrous: assigning a worker necessarily excludes others. Thus, in any efficient equilibrium, workers’ effort incentives when assigned are dynamic, arising from an expectation that their future workload upon a current good output is sufficiently lighter than that upon a bad one, and these future assignments must be structured in a way that preserves other workers’ effort incentives.

Our main result shows that all uniformly efficient assignment rules take the form of “first-order rotation.” In each period, the manager assigns the lowest ranked worker to the task. A bad output leaves the ranking unchanged. A good output promotes the incumbent to the top rank and shifts others down by one rank. Thus, each worker, when assigned, is retained after a bad output and is relieved after a good output until all other workers have been assigned and delivered a good output. First-order rotation rules are unique up to workers’ initial ranking and therefore provide identical dynamic incentives.

We show that this result, and its subsequent implications, extend naturally to settings where the task is desirable for the workers, where tasks vary over time, and where the manager repeatedly assigns multiple concurrent tasks. With a desirable task, rotation is reversed: the incumbent is relieved after a bad output but not a good one since assignment serves as a reward rather than punishment. We also show that our results extend when the manager has commitment power, when workers face private payoff shocks, and when workers have limited information—they may observe neither the assignee’s identity nor his output unless they themselves are assigned, and may be uncertain about the workforce size.

An important implication of our results is that efficiency conflicts with equality. We show that in any efficient equilibrium corresponding to first-order rotation, workers’ continuation payoffs are strictly ranked in every period, reflecting that their average workloads are strictly ranked in every continuation. This holds even for workers who are independent and symmetric in all aspects. Indeed, efficiency alone leads to this tension: we show that in any efficient

equilibrium, in each period on path, workers’ continuation payoffs cannot all be equal; the uniformity criterion merely sharpens this tension by forcing a strict payoff ranking.

In practice, nonmonetary discrimination plays a substantial role in labor market inequality.¹ Unequal workload, in particular, may be attributed to worker heterogeneity—differences in production costs (Board, 2011), productivity (Andrews and Barron, 2016), or demographic characteristics.² Our results complement these accounts by suggesting that unequal work assignment may reflect optimal incentive provision rather than worker heterogeneity. Our results also highlight that the efficiency-equality tension is more pervasive than previously recognized. With independent and symmetric workers, this tension does not arise in monetary contracting models (Holmström, 1982). With complementary and symmetric workers, it arises in the context of static, public monetary contracts (Winter, 2004; Halac et al., 2021). The tension here is dynamic, nonmonetary, concerns independent workers and, as we show in our extension, also robust to workers’ uncertainty about how other workers are treated.

Another important implication of our results is that the scope of efficiency is governed by the ability of first-order rotation in sustaining efficiency. We leverage this to study organizational design. We show that improving shirking detection, increasing workforce size, and widening the payoff gap between assignment and unassignment all expand the scope of efficiency. In contrast, improving effort detection may perversely reduce it, unlike a standard prediction that better information facilitates cooperation (Kandori, 1992). We further show that instruments to improve shirking detection are robustly superior relative to other instruments for organizations seeking to promote efficiency without exacerbating inequality, measured by the differences in workers’ continuation payoffs across ranks under first-order rotation. Specifically, we show that non-information instruments—workforce size and the assignment-unassignment payoff gap—necessarily increase inequality to various extent as they expand the scope of efficiency, while improving shirking detection expands the scope of efficiency without affecting inequality. Moreover, improving shirking detection robustly dominates varying effort detection, which may increase inequality while promoting efficiency.

Methodologically, our analysis overcomes a difficulty in repeated principal-multi-agent games without transfers. Our results involve characterizing, for any number of workers, the largest set of parameters for which an efficient equilibrium exists. As noted in De Clippel,

¹See, e.g., Adi Gaskell, “How Non-financial rewards widen workplace inequality,” *Forbes*, July 7, 2021.

²See, e.g., Joan C. Williams and Marina Multhaup, “For women and minorities to get ahead, managers must assign work fairly,” *Harvard Business Review*, March 5, 2018.

Eliaz, Fershtman and Rozen (2021), such characterization has not been obtained in the literature as the equilibrium payoff set is not well understood, making it difficult to identify its extreme points that determine necessary conditions for efficiency. We circumvent this difficulty by proposing a dual characterization approach, simultaneously solving for the boundary of the largest set of parameters for which an efficient equilibrium exists and the associated efficient equilibrium payoffs. This boundary yields the desired set of parameters.

While we focus on performance-contingent task assignment to address moral hazard, practices where tasks are assigned without regard to performance are also common.³ For instance, shifts in hospitals are often fixed in advance. Our results do not dispute these practices. Rather, by showing that uniform efficiency results in inequality among workers, our results suggest that the prevalence of fixed assignment schemes may reflect an organizational preference for fairness, morale, or institutional norms from which our model abstracts.

Related literature. We contribute to the literature on dynamic multi-agent assignment. Our model is most closely related to Board (2011) and Andrews and Barron (2016) who study assignment under dynamic moral hazard but with flexible monetary transfers. Their assignment dynamics are driven by observable, time-varying worker heterogeneity and transfers; both features are absent in our model. Further, the dynamics of Board (2011) and Andrews and Barron (2016), unlike ours, feature a form of favoritism where the manager tends to retain a worker until a significant shock regarding workers’ heterogeneity occurs. De Clippel et al. (2021) study assignment among ex ante identical workers without monetary transfers, but under dynamic adverse selection rather than moral hazard. They focus on assignment rules that treat unassigned workers symmetrically in each period; in our model, symmetric treatment of unassigned workers undermines efficiency.

Our analysis contrasts with existing work on dynamic single-agent assignment (Li, Matouschek and Powell, 2017; Bird and Frug, 2019; Guo and Hörner, 2020; Lipnowski and Ramos, 2020). In these models, assignment is not rivalrous: the principal decides whether or not to assign, but not whom to assign. Our analysis also contrasts with models of dynamic moral hazard in teams (Che and Yoo, 2001; Rayo, 2007; Bonatti and Hörner, 2011; Georgiadis, 2015). In these models, workers are complementary and there are no assignment dynamics.

³There is a literature on non-performance-contingent task assignment (e.g., Ickes and Samuelson, 1987; Meyer, 1994; Ortega, 2001; Arya and Mittendorf, 2004; Arya and Mittendorf, 2006a,b; Prescott and Townsend, 2006; Hakenes and Katolnik, 2017; Eliaz, Fershtman and Frug, 2025).

2 Baseline model

We begin with a baseline model before extending it in various directions in Section 7. Time $t = 0, 1, \dots$ is discrete and the horizon is infinite. There is a manager and a set of ex ante identical workers $N := \{1, \dots, n\}$, where $n \geq 1$. The manager is endowed with a task that is undesirable for the workers.

In each period the manager publicly assigns one worker to perform the task.⁴ The assigned worker privately chooses whether to exert effort or to shirk. Effort yields a good output $\bar{y}$ with probability $p \in (0, 1)$ and a bad output $\underline{y}$ otherwise; shirking yields a good output with probability $q \in (0, p)$ and a bad output otherwise. This output is a public noisy signal of the worker's action. The next period then unfolds.

In each period t , each worker i 's payoff, denoted by u_t^i , and the manager's payoff, denoted by v_t , are as follows. The assigned worker gets payoff normalized to zero if he exerts effort and payoff $s > 0$ if he shirks. Each unassigned worker receives payoff $r > s$. Therefore, workers prefer to be unassigned than assigned;⁵ in addition, when assigned, they prefer to shirk than to exert effort. The manager's payoff depends only on the output y and is given by some function $g(y)$, with $g(\underline{y}) < 0 < g(\bar{y})$ so that she prefers a good output to a bad one.

A public history in each period t , denoted by h_t , is an element in $(N \times \{\bar{y}, \underline{y}\})^t$, consisting of the identities of previously assigned workers and their outputs, with $h_0 := \emptyset$. Worker i 's strategy is a collection $\sigma^i \equiv (\sigma_t^i)_{t=0}^\infty$, where $\sigma_t^i(h_t) \in [0, 1]$ specifies his probability of exerting effort in period t upon being assigned at history h_t . Note that the worker does not condition his strategy on his past private actions; this restriction is innocuous for our results, because monitoring has a product structure.⁶ The manager's strategy is a collection $f \equiv (f_t)_{t=0}^\infty$, where $f_t(h_t) \in \Delta(N)$ specifies a distribution over workers from which the currently assigned

⁴This rules out the possibility that the manager assigns no worker. This is innocuous for our results: we shall focus on efficient equilibria and, to the extent that efficiency is concerned, the manager must assign a worker in each period.

⁵Our analysis goes through by simply assuming $r > 0$, namely not ranking workers' preference between being unassigned and shirking. It is nonetheless an implication of our results that achieving efficiency in equilibrium requires $r > s$.

⁶Fudenberg and Levine (1994) show that in repeated games with imperfect public product-structure monitoring, such as ours, the set of sequential equilibrium payoffs and the set of perfect public equilibrium payoffs coincide. Thus, to the extent that we are concerned with efficient equilibrium payoffs, there is no loss of generality in restricting attention to public strategies (and using perfect public equilibrium as the solution concept, as we do). There is also no loss of generality for equilibrium behavior: in any efficient equilibrium, workers exert effort when assigned on path, irrespective of whether attention is restricted to public strategies.

worker is drawn at history h_t .

The workers and the manager share a common discount factor $\delta \in (0, 1)$.⁷ Each worker i 's and the manager's average realized payoffs are

$$(1 - \delta) \sum_{t=0}^{\infty} \delta^t u_t^i \quad \text{and} \quad (1 - \delta) \sum_{t=0}^{\infty} \delta^t v_t.$$

We use perfect public equilibrium (Fudenberg, Levine and Maskin, 1994), henceforth equilibrium, as the solution concept. An equilibrium exists: repeated play of static Nash equilibrium, such as one in which workers shirk when assigned and the manager assigns worker 1, is an equilibrium. This equilibrium is fully inefficient.

We assume that effort is efficient, namely that the social surplus associated with effort is at least its counterpart associated with shirking:

$$pg(\bar{y}) + (1 - p)g(\underline{y}) + (n - 1)r \geq qg(\bar{y}) + (1 - q)g(\underline{y}) + s + (n - 1)r. \quad (1)$$

An equilibrium is efficient if each worker exerts effort when assigned (on path), and efficient equilibria attain the manager's first-best payoff.⁸ We interpret the manager's strategy as an assignment rule, and refer to the manager's strategy in any efficient equilibrium as an efficient assignment rule. Note that the manager's strategy is flexible: her assignment in each period possibly depends on past assignments and past outputs in arbitrary ways.

Our interest is in efficient assignment rules that are not parameter-sensitive, as motivated in the Introduction. Formally, we introduce a robustness criterion that precludes these rules in Definition 1 below. For any number n of workers, let

$$\Omega(n) := \{(p, q, r, s, \delta) \in (0, 1)^2 \times \mathbf{R}_{++}^2 \times (0, 1) : p > q\} \quad (2)$$

be the set of feasible parameter vectors, with typical element ω , and let $\Omega^*(n)$ be the largest set of such parameter vectors for which an efficient equilibrium exists. We call each $\omega \in \Omega^*(n)$ an efficient parameter vector.

⁷Our results go through if the manager has a different discount factor.

⁸Our analysis of efficient equilibria is unlike in canonical repeated partnership games where strong symmetric strategies are inefficient (e.g., Radner, Myerson and Maskin, 1986). In our model, efficiency requires that workers play strongly symmetric strategies (on path)—exerting effort when assigned—in equilibrium.

Definition 1 (Uniform efficiency). *For each n , an assignment rule is said to be uniformly efficient if it is efficient simultaneously for all $\omega \in \Omega^*(n)$.*

We focus on efficient equilibria corresponding to uniformly efficient assignment rules. *A priori*, a uniformly efficient assignment rule need not exist; our main result shows that it does and characterizes all these rules. Importantly, the uniformity criterion entails no loss of generality for efficiency: by definition, a uniformly efficient assignment rule constitutes an efficient equilibrium whenever such an equilibrium exists. Therefore, no assignment rule outperforms any uniformly efficient assignment rule in terms of achieving efficiency.

The uniformity criterion is different from the Blackwell criterion, another robustness criterion that was introduced by [Blackwell \(1962\)](#) in dynamic programming and adapted by [Cavounidis, Ghosh, Hörner, Solan and Takahashi \(2025\)](#) in the context of repeated games. The Blackwell criterion requires strategies to be efficient or optimal for sufficiently high discount factors simultaneously. Uniformity is stronger, demanding simultaneous efficiency across all parameters for which efficiency is attainable. Uniformly efficient assignment rules are Blackwell efficient, but the converse is false.

Because workers' actions are hidden, only the manager possibly has an observable deviation in equilibrium. For concreteness, we assume that following any off-path history, workers shirk when assigned and the manager assigns worker 1. Our results do not hinge on this assumption: along the path of any efficient equilibrium, the manager attains her first-best (expected) payoff $pg(\bar{y}) + (1 - p)g(\underline{y})$ in each period and has no profitable deviation; in particular, our assumption on off-path behavior is relevant for whether the manager has a profitable deviation. This observation also implies that granting the manager commitment power so that she commits to an assignment rule at the outset does not change our results. Hereafter, we omit mentioning off-path behavior for conciseness and focus on workers' incentives.

3 Effort incentives

In this section, we present a useful preliminary result formalizing workers' incentive structure. In any equilibrium, denote worker i 's continuation payoff at history h_t by

$$U^i(h_t) := \mathbf{E} \left[(1 - \delta) \sum_{\tau=t}^{\infty} \delta^{t-\tau} u_{\tau}^i \middle| h_t \right],$$

where the expectation is taken with respect to the equilibrium outcome distribution. Let $h_t i y$ be the public history that is a concatenation of history h_t followed by an assignment of worker i who then produces output y .

Lemma 1 (Incentives). *In any efficient equilibrium, at any history h_t on path where worker i is assigned with positive probability,*

$$U^i(h_t i \bar{y}) - U^i(h_t i y) \geq \frac{(1 - \delta)s}{\delta(p - q)}, \quad (3)$$

and

$$\sum_{j=1}^n U^j(h_t) = (n - 1)r. \quad (4)$$

The proofs of all formal results are in Appendix A. The inequality (3) formalizes that effort incentives constrain future assignments in efficient equilibrium. To interpret this, note first that effort incentives are intrinsically dynamic: if the interaction were one-shot, then each worker would have a strict incentive to shirk upon being assigned; therefore, a worker is willing to exert effort when assigned only if his continuation payoff upon a good output sufficiently exceeds that upon a bad output, namely (3) holds. Because these continuation payoffs are determined by future assignments, any efficient assignment rule that sustains this inequality must relieve the incumbent worker from sufficiently many future assignments in the continuation upon a good output while promising him a sufficiently heavy assignment burden in the continuation upon a bad output. The incentive constraint (3) further implies that for an efficient equilibrium to exist, there must be at least two workers, which we assume hereafter. If there were only one worker, then this worker must always be assigned and receive zero payoff, contradicting (3).

The equality (4) states that the workers' continuation payoffs, and hence the manager's incentive provision, are bound by the rivalrous nature of task assignment: in any efficient equilibrium, in each period (on path), raising one worker's continuation payoff necessarily reduces at least one other worker's continuation payoff. This constraint holds because in every period, the manager's assignment of any one worker excludes the others so that the assigned worker obtains stage payoff 0 while the other $n - 1$ workers obtain stage payoff r .

Before proceeding, we remark that the incentive structure in the case of three or more

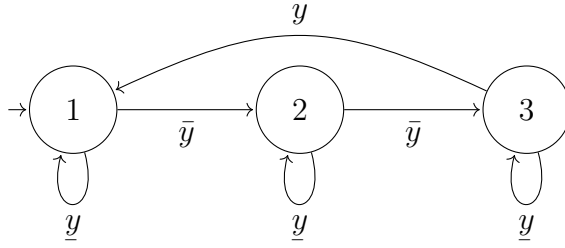


Figure 1: First-order rotation with three workers

workers is fundamentally richer than that of the two-worker case. The former case differs from the latter in that granting respite to one worker requires the manager to choose a replacement from a pool of (at least two) unassigned workers; this choice affects workers' continuation payoffs and incentives. Accordingly, in all subsequent examples that we use to illustrate our results, we consider the case of three workers rather than two.

4 Uniformly efficient assignment

In this section, we report our main result. Definition 2 is essential.

Definition 2 (First-order rotation). *An assignment rule is a first-order rotation rule if it satisfies the following. In each period, workers are distinctly labeled $\pi_1, \dots, \pi_n$ and the worker currently labeled π_n is assigned. Upon a bad output, all workers retain their current labels. Upon a good output, worker currently labeled π_k is relabeled $\pi_{(k \bmod n)+1}$, for each $k = 1, \dots, n$.*

Under any first-order rotation rule, in each period, the manager strictly ranks all workers; for each k , worker labeled π_k denotes the k th ranked worker, and higher index k corresponds to lower rank. The manager assigns the lowest ranked worker to the task. If the output turns out to be bad, the ranking is unchanged. If the output turns out to be good instead, the lowest ranked worker becomes top ranked while all other workers are shifted down by one rank. Consequently, the manager rotates the task through all workers based only on first-order information—the incumbent's identity and his current output, retaining him after a bad output and relieving him after a good output until all other workers have been assigned and delivered a good output. Note that first-order rotation rules utilize all workers, although the manager has the option to permanently bench some worker. Figure 1 illustrates first-order rotation in the case of three workers, where a lower-indexed worker has a lower initial rank.

By definition, first-order rotation rules are unique up to workers' initial ranking. Because initial rankings do not affect workers' (dynamic) effort incentives, all first-order rotation rules provide identical incentives. In the efficient equilibrium corresponding to any first-order rotation rule, at every history on path, the continuation payoff of a worker is a function of his rank π , which we denote as $U(\pi)$, and is identical to its counterpart in the efficient equilibrium corresponding to any other first-order rotation rule. Hereafter, when no risk of ambiguity arises, we refer to the efficient equilibrium corresponding to any first-order rotation rule as the efficient equilibrium under first-order rotation.

Despite the simplicity of first-order rotation and the manager's freedom to condition assignment on the entire history of assignments and outputs, our main result shows that first-order rotation rules are the unique class of assignment rules that are uniformly efficient:

Theorem 1 (Characterization). *An assignment rule is uniformly efficient if and only if it is a first-order rotation rule.*

The content of Theorem 1 is threefold. First, uniformly efficient assignment rules exist, since first-order rotation rules exist by construction.

Second, first-order rotation rules sustain efficient equilibria whenever an efficient equilibrium exists. Therefore no assignment rule outperforms first-order rotation in achieving efficiency. This includes rules with probabilistic rewards despite their prominence in the repeated games literature,⁹ as well as rules that utilize workers' relative performance despite the competitive nature of the rivalrous environment.¹⁰ Intuitively, because equilibrium has a recursive structure and workers are symmetric, maximal moral hazard that permits efficiency—corresponding to parameter vectors on the boundary of the largest set of efficient parameter vectors—forces identical incentives to be provided to all workers whenever they are assigned, and also requires these incentives to be extreme. The harshest feasible punishment

⁹See Appendix B.1 for an illustration. In standard repeated games with imperfect monitoring, probabilistic transitions are often used to limit time spent in inefficient phases while preserving incentives. Here, all continuations in efficient equilibria are efficient. Probabilistic transitions then offer no advantage over deterministic ones and only weaken incentives.

¹⁰Relative performance information is pure noise relative to the worker's current effort that the manager seeks to motivate in each period. Therefore, the disregard of this information in uniformly efficient assignment rules echoes the informativeness principle in standard monetary contracting models (Holmström, 1979, 1982). For comparison, here is an example of an assignment rule that utilizes workers' relative performance. In each period, the manager computes, for each worker, a score equal to the number of good outputs minus that of bad outputs this worker previously produced, and then assigns the worker who has the lowest score, breaking ties uniformly at random.

is immediate retention, while the strongest feasible reward is relief until every other worker has been assigned and delivered a good output; under the rivalry constraint, no stronger reward that is identical across workers can be given. Since these incentives work when moral hazard is maximal while permitting efficiency, they work when moral hazard is less severe.

Third, for some efficient parameter vectors, first-order rotation is necessary for sustaining an efficient equilibrium. Indeed, it can be shown that for each workforce size n , an efficient assignment rule must prescribe first-order rotation if and only if the parameter vector lies on the boundary of the largest set of efficient parameter vectors $\Omega^*(n)$. The “if” direction is established in the proof of Theorem 1, reflecting the discussion in the previous paragraph. For the “only if” direction, it is easy to verify that for each parameter vector on the interior of $\Omega^*(n)$, the assignment rule we construct in Appendix B.1 sustains efficiency with parameter-sensitive, probabilistic rewards, so that first-order rotation is not necessary for efficiency.

In the rest of this section, we sketch a proof of Theorem 1. For each workforce size $n \geq 2$ and parameter vector $\omega \in \Omega(n)$, let $E \subseteq \mathbf{R}_+^n$ be the set of workers’ equilibrium payoff vectors,¹¹ with typical element $U = (U^1, \dots, U^n)$ where U^i denotes worker i ’s payoff. Let E^* be the set of efficient equilibrium worker payoff vectors. Standard arguments show that E is compact (Abreu, Pearce and Stacchetti, 1990), and so is E^* , since $E^* = \{U \in E : \sum_{i=1}^n U^i = (n-1)r\}$ by the rivalry constraint (4). We construct an efficient equilibrium payoff vector $U^* := (\underline{U}, U^{(1)}, \dots, U^{(n-1)})$. Here, $\underline{U} := \min_{(U^1, \dots, U^n) \in E^*} U^1$ denotes the lowest payoff a worker can obtain in any efficient equilibrium; $U^{(1)} := \max_{(\underline{U}, U^2, \dots, U^n) \in E^*} U^2$ denotes the highest payoff a worker can obtain conditional on some other worker obtaining $\underline{U}$ in any efficient equilibrium; for each $k = 2, \dots, n-1$, $U^{(k)} := \max_{(\underline{U}, U^{(1)}, \dots, U^{(k-1)}, U^{k+1}, \dots) \in E^*} U^{k+1}$ denotes the highest payoff a worker can obtain conditional on k other workers obtaining $\underline{U}, U^{(1)}, \dots, U^{(k-1)}$ in any efficient equilibrium. Finally, define $\bar{U} := \max_{(U^1, \dots, U^n) \in E^*} U^1$ as the highest payoff a worker can obtain in any efficient equilibrium. We write E^* as E_ω^* to stress its dependence on ω .

Fix $n \geq 2$. The rest of the proof proceeds in three steps. The first step characterizes the largest set of efficient parameter vectors, $\Omega^*(n)$. This set is closed and hence contains its boundary that we denote by $\text{bd}(\Omega^*(n))$, because it consists of parameter vectors for which an efficient equilibrium exists where the incentive constraint (3)—a weak inequality—holds for each worker when assigned. To characterize $\Omega^*(n)$, we solve simultaneously for the boundary

¹¹Here we omit the manager’s payoff. As mentioned, in any efficient equilibrium, after any history (on path), the manager’s continuation payoff is equal to $pg(\bar{y}) + (1-p)g(y)$.

$\text{bd}(\Omega^*(n))$ and the set E_ω^* for each parameter vector $\omega \in \text{bd}(\Omega^*(n))$. For each such parameter vector, moral hazard is maximally tight so that the payoff vector U^* we construct above is decomposed, in the sense of [Abreu et al. \(1990\)](#), as follows:

$$\begin{pmatrix} \underline{U} \\ U^{(1)} \\ \vdots \\ U^{(n-1)} \end{pmatrix} = (1 - \delta) \begin{pmatrix} 0 \\ r \\ \vdots \\ r \end{pmatrix} + \delta \left[p \begin{pmatrix} U^{(1)} \\ U^{(2)} \\ \vdots \\ \underline{U} \end{pmatrix} + (1 - p) \begin{pmatrix} \underline{U} \\ U^{(1)} \\ \vdots \\ U^{(n-1)} \end{pmatrix} \right]. \quad (5)$$

The first row concerns a currently assigned worker 1, stating that his continuation payoff is $\underline{U}$, with a good output changing this payoff to $U^{(1)}$ and a bad output leaving it unchanged. Intuitively, because moral hazard is maximally tight, his incentive constraint (3) binds even when his payoff is $\bar{U}$ upon a good output and is $\underline{U}$ upon a bad output, namely

$$\bar{U} - \underline{U} = \frac{(1 - \delta)s}{\delta(p - q)}, \quad (6)$$

and worker 1's current continuation payoff attains the lower bound $\underline{U}$. By definition of $U^{(1)}$, it then follows that $\bar{U} = U^{(1)}$ because in each period, by symmetry, the assigned worker receives $\underline{U}$; this establishes the first row. The second row concerns an unassigned worker 2, stating that his current continuation payoff is $U^{(1)}$, with a good output by worker 1 changing his payoff to $U^{(2)}$ while a bad output leaving it unchanged. This holds because given worker 1's payoff evolution, any alternative worker 2's payoff evolution makes him worse off, contradicting the definition of $U^{(1)}$. The other rows follow analogously. The system (5) has a unique solution U^* , which we denote by $(V^k(n, p, r, \delta))_{k=1}^n$. By symmetry of the workers, the boundary $\text{bd}(\Omega^*(n))$ is characterized by all feasible parameter vectors that satisfy the binding incentive constraint (6), with $\underline{U} = V^1(n, p, r, \delta)$ and $U^{(1)} = V^2(n, p, r, \delta)$. This yields the desired set of parameter vectors:

$$\Omega^*(n) = \left\{ \omega \in \Omega(n) : V^2(n, p, r, \delta) - V^1(n, p, r, \delta) \geq \frac{(1 - \delta)s}{\delta(p - q)} \right\}. \quad (7)$$

The second step shows that all first-order rotation rules are uniformly efficient. Given any first-order rotation rule and efficient parameter vector ω , in the efficient equilibrium corresponding to this rule, the workers' continuation payoff vector in any period is given by, up

to permutation of entries, $U^* = (\underline{U}, U^{(1)}, \dots, U^{(n-1)})$ characterized in (5). If $\omega \in \text{bd}(\Omega^*(n))$, the first step has highlighted that each worker's incentive constraint for effort when assigned holds (with equality). The proof then verifies that this incentive constraint holds if ω lies on the interior of $\Omega^*(n)$ where the moral hazard problem is less severe. Hence all first-order rotation rules are efficient over all $\omega \in \Omega^*(n)$, i.e., they are uniformly efficient.

The final step establishes that uniformly efficient assignment rules must prescribe first-order rotation. Because any such rule is efficient on $\text{bd}(\Omega^*(n))$ by definition, it suffices to show that given any such parameter vector on this boundary, every efficient equilibrium prescribes first-order rotation. We use our construction in the first step to show that any currently assigned worker's continuation payoff must cycle through the sequence $\underline{U}, U^{(1)}, \dots, U^{(n-1)}$, changing value after a good output but staying put after a bad output. This implies that each currently assigned worker is retained after a bad output and is relieved after a good output until all other workers have been assigned and have produced a good output, allowing us to conclude that the manager's strategy must prescribe first-order rotation.

5 Efficiency and inequality

In this section, we show that efficient task assignment conflicts with equality and explain why this tension differs from standard incentive models. Recall that in the efficient equilibrium under first-order rotation, in each period on path, the continuation payoff of a worker with rank π is denoted by $U(\pi)$.

Proposition 1 (Efficiency-equality tension). *In the efficient equilibrium under first-order rotation, $U(\pi_k)$ is strictly decreasing in k .*

Proposition 1 shows that in the efficient equilibrium under first-order rotation, higher ranked workers have strictly higher continuation payoffs in each period. This is intuitive, as higher ranked workers enjoy longer respite. Therefore, despite being ex ante identical and playing identical strategies, workers are strictly discriminated against each other in every period on path, including those who have been assigned for the same number of times and have identical output history. This implies that in every continuation, workers' expected average workloads, namely their average discounted frequencies of being assigned, are strictly ranked. Because uniform efficiency prescribes first-order rotation by Theorem 1, Proposition 1

implies that uniform efficiency necessarily leads to discrimination in every continuation.¹²

In fact, efficiency alone demands discrimination; the uniformity criterion only sharpens it by forcing a strict workload ranking. More precisely, in any efficient equilibrium, at each history h_t on path, worker i 's incentive constraint (3) when assigned requires that his continuation payoffs upon good and bad outputs, $U^i(h_t i \bar{y})$ and $U^i(h_t i \underline{y})$, satisfy $U^i(h_t i \bar{y}) > U^i(h_t i \underline{y})$. If there were no discrimination so that all workers' continuation payoffs are equal at each of these continuations, the rivalry constraint (4) would imply $U^i(h_t i \bar{y}) = U^i(h_t i \underline{y}) = (n-1)r/n$, leading to a contradiction. Thus, efficiency requires that workers who are assigned the task early are not perfectly compensated with a lower future workload, ruling out equality.

For illustration, Figure 2 depicts an assignment rule where unassigned workers are treated equally, just as in [De Clippel et al. \(2021\)](#), in the case of three workers. Specifically, worker 1 is initially assigned and in each period, if worker i is assigned, then upon a good output, the next worker is assigned among the two workers other than i with equal probability; upon a bad output, worker i is retained. This rule is efficient over certain parameter vectors—those in which the shirking gain s is sufficiently small—because it grants workers respite after good outputs, but there exist parameter vectors under which this assignment rule is not efficient while first-order rotation rules are in view of Theorem 1. It can be readily verified that in the efficient equilibrium corresponding to this rule, at every history on path, there is discrimination: the two unassigned workers' continuation payoffs are identical and are strictly higher than the assigned worker's one.

Proposition 1 highlights that the efficiency-equality tension is more pervasive than previously recognized. Standard monetary contracting models predict that ex ante identical and technologically independent workers are treated equally. Specifically, in those models, the informativeness principle ([Holmström, 1979, 1982](#)) implies that at the optimum, a manager contracts independently with workers who are technologically independent, and therefore symmetrically if these workers are in addition symmetric. In our model, the rivalry of assignment prevents independent and symmetric treatment of workers in efficient equilibria: granting a worker respite requires assigning another; efficiency then demands discrimination. The seminal work of [Winter \(2004\)](#) identifies an efficiency-equality tension in the context of static monetary contracting with technologically complementary workers, and [Halac et al. \(2021\)](#)

¹²Of course, there is a first-order rotation rule that assigns the initial ranking uniformly at random so that workers' ex ante payoffs are identical, just like in [Winter \(2004\)](#); nonetheless, discrimination occurs in every period once the initial ranking is realized.

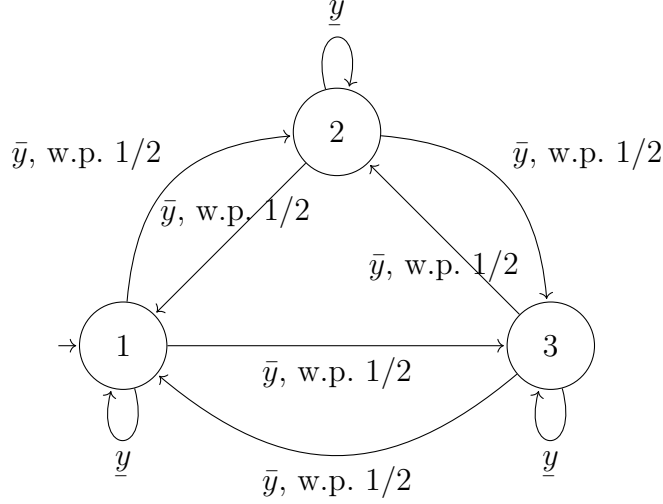


Figure 2: Symmetric treatment of unassigned workers

show that this tension in [Winter \(2004\)](#) hinges on public worker ranks. The discrimination highlighted by Proposition 1 transcends these dimensions. It is nonmonetary, pertaining to assignment priority. It is dynamic, as rankings evolve over time. It concerns technologically independent rather than complementary workers. Finally, as we elaborate in Section 7, this discrimination is also robust to rank uncertainty, in the sense that Theorem 1 and Proposition 1 extend even if workers are uncertain about their relative ranks under first-order rotation.

6 Organizational design

In this section, we interpret several model parameters as organizational design instruments and study how they affect the scope of efficiency, namely whether an efficient equilibrium exists. We distinguish between non-information instruments, namely workforce size n and the assignment-unassignment payoff gap r , as well as information instruments, namely effort detection p and shirking detection q . For instance, organizations can affect n through staffing decisions, r by adjusting off-duty benefits or working conditions, p through improved machinery that affects the productivity of effort but not shirking, and q through activity logs that affect shirking detection without affecting the productivity of effort.

By Theorem 1, an efficient equilibrium exists if and only if first-order rotation is efficient. In efficient equilibria under first-order rotation, we write workers' continuation payoff $U(\cdot)$

as $U(\cdot; n, p, r)$ to stress its dependence on the instruments; it is independent of q because workers do not shirk on path. First-order rotation rules are efficient if and only if workers' incentive constraint (3) holds:

$$S(n, p, q, r) := U(\pi_1; n, p, r) - U(\pi_n; n, p, r) - \frac{(1 - \delta)s}{\delta(p - q)} \geq 0, \quad (8)$$

where each worker, when assigned, obtains continuation payoff $U(\pi_1; n, p, r)$ upon a good output and $U(\pi_n; n, p, r)$ upon a bad output. We call $S(n, p, q, r)$ the scope of efficiency.

Proposition 2 (The scope of efficiency). *The following hold.*

1. $S(n, p, q, r)$ is strictly increasing in n and r , and is strictly decreasing in q .
2. There exists $p^* \in (0, 1]$ such that $S(n, p, q, r)$ is strictly increasing in p on (q, p^*) and strictly decreasing in p on $[p^*, 1]$. There exists $\underline{r} > 0$ such that $p^* \in (0, 1)$ if $r > \underline{r}$ and $p^* = 1$ otherwise.

Part 1 shows that improving shirking detection, increasing workforce size, and widening the assignment-unassignment payoff gap all unambiguously expand the scope of efficiency. Intuitively, under first-order rotation, workers' effort incentives are stronger if shirking is less likely to yield a good output, the duration of respite following success is longer, and workers have stronger preference between being assigned and being unassigned. This further highlights that our model is unlike standard competitive environments where the discouragement effect is present and predicts that a larger pool of participants undermines effort (Nalebuff and Stiglitz, 1983, Taylor, 1995); indeed, first-order rotation subverts rivalry rather than promoting competition by disregarding workers' relative performance.

Part 2 shows that improving effort detection has, in general, ambiguous effects on the scope of efficiency and may even reduce it unless effort detection is initially poor or the assignment–nonassignment payoff gap is small. Intuitively, better effort detection has two countervailing effects. On the one hand, it strengthens incentives by making effort more likely to generate success. On the other hand, because the improvement applies to all workers, it accelerates rotation across ranks and shortens respite, weakening incentives. When the payoff gap is small and effort detection is initially poor so that rotation is slow, the positive effect dominates; otherwise the negative effect can prevail. This potential perverse outcome due to better effort detection contrasts with canonical long-run relationships where such improvement does not reduce the set of equilibrium payoffs (Kandori, 1992).

Unequal worker treatment might be detrimental to morale and retention in organizations. While such considerations lie outside our model, we next examine how the various instruments that expand the scope of efficiency affect inequality under first-order rotation. We measure the inequality between any two workers ranked k th and k' th by their (absolute) payoff difference

$$I(k, k'; n, p, r) := |U(\pi_k; n, p, r) - U(\pi_{k'}; n, p, r)|$$

in efficient equilibrium under first-order rotation.¹³ Note that $I(k, k'; n, p, r)$ is symmetric in k and k' . We call $I(1, n; n, p, r)$, or equivalently $I(n, 1; n, p, r)$, the scope of inequality, as this is the maximum difference between the continuation payoffs associated with any two ranks by Proposition 1.

Proposition 3 (Inequality). *Fix distinct ranks $k \neq k'$. Then:*

1. $I(k, k'; n, p, r)$ is independent of q .
2. $I(k, k'; n, p, r)$ is strictly decreasing in n . An exception arises when one rank is n : $I(1, n; n, p, r) = I(n, 1; n, p, r)$ is strictly increasing in n .
3. $I(k, k'; n, p, r)$ is strictly increasing in r .
4. If one of k, k' equals n , then $I(k, k'; n, p, r)$ is strictly decreasing in p . If neither equals n , there exists $p^* \in (0, 1)$ such that $I(k, k'; n, p, r)$ is strictly increasing in p on $(0, p^*)$ and strictly decreasing in p on $[p^*, 1)$.¹⁴

Part 1 shows that shirking detection does not affect worker inequality. This is because workers exert effort when assigned in efficient equilibria on path. Part 2 shows that increasing workforce size reduces inequality between any fixed rank pair but widens the gap between the top and bottom ranks. This is because a larger workforce lengthens respite after success, increasing the spread between extreme ranks, while reducing marginal gains from moving up the ranking. Part 3 shows that expanding the assignment-unassignment payoff gap for any fixed rank pair, as it scales the value of respite across all ranks. Finally, part 4 shows that improving effort detection unambiguously reduces inequality between any rank pair involving the bottom rank but have ambiguous effects on inequality between other rank pairs.

¹³Adopting any other measure of inequality that is a positive transformation of this payoff difference does not affect our results.

¹⁴For course, there are values of p that fall below p^* only if q is sufficiently small since $p > q$.

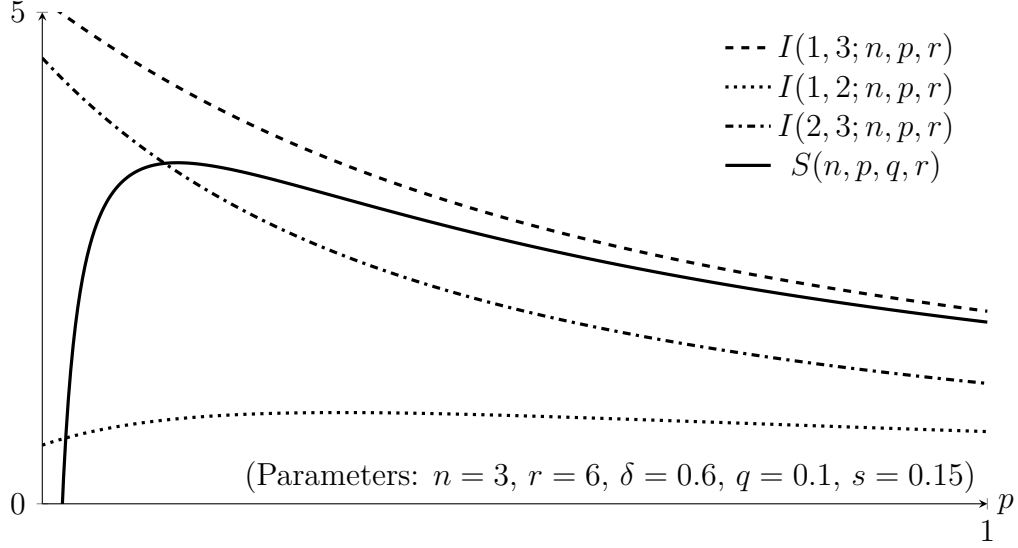


Figure 3: Scope of efficiency and inequalities

Intuitively, the resulting faster rank transitions reduce inequality by reducing the time spent at any rank but also increase workers' value of being assigned later in the future, which increases inequality. This latter effect arises only among non-bottom-ranked workers who are not currently assigned, whereas the former reduction applies across all ranks. Therefore, inequality between any rank pair involving the bottom rank falls. For other rank pairs, inequality falls if and only if the current quality of effort detection is sufficiently high, so that current rank rotation is sufficiently fast. Figure 3 illustrates.

Taken together, Proposition 2 and Proposition 3 suggest that improving shirking detection is particularly attractive for organizations seeking to expand the scope of efficient incentive provision without exacerbating inequality. It is superior to noninformation instruments because it expands the scope of efficiency without affecting inequality. Its comparison with improving effort detection is more nuanced. Indeed, improving output precision can, for some parameters, expand efficiency while reducing inequality—the latter outcome occurs when workers are sufficiently patient and current effort detection is sufficiently poor; Figure 4 illustrates this case, and Appendix B.2 formalizes it. This asymmetry underscores the robustness advantage of policies that target shirking detection.

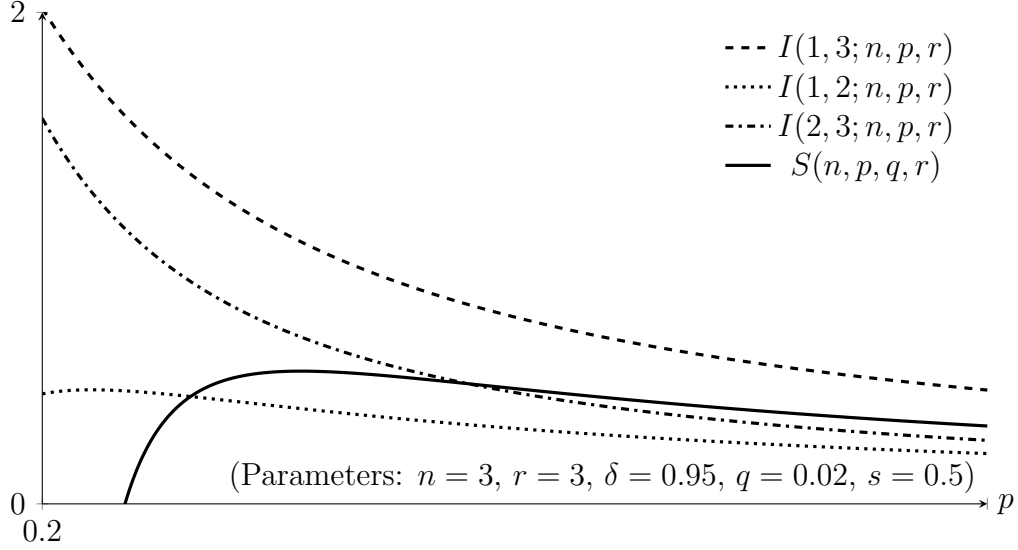


Figure 4: Scope of efficiency and inequalities

7 Extensions

In this section, we explain that our results extend in several alternative versions of our baseline model.

Idiosyncratic effort cost. Consider a version of our baseline model in which the workers face private idiosyncratic shocks to their opportunity cost to exert effort. Specifically, suppose that in each period, each worker i has a private shirking gain s^i that is independently drawn across time and workers from some (possibly worker-specific) distribution on a nondegenerate compact interval $[0, \bar{s}]$ for some $\bar{s} > 0$. Suppose that efficiency continues to require that a worker exerts effort whenever assigned, namely (1) holds with s replaced by $\bar{s}$. Our results extend, with the difference that s is replaced by $\bar{s}$ in the expression for the largest set of efficient parameters in (7): if an efficient equilibrium exists, then at every history, the assigned worker must have a best reply to exert effort irrespective of his realized shirking gain.

Desirable task. Consider a version of our baseline model where the manager's task is desirable for the workers so that workers prefer to be assigned. Specifically, suppose that in each period, the unassigned worker gets payoff $-r < 0$ instead of r . The model is otherwise unchanged. In particular, upon being assigned, each worker continues to prefer to shirk than to exert effort, and higher r continues to represent stronger worker preference between

being assigned and being unassigned. Almost all our results extend, with essentially identical arguments except that assignment here serves as a reward rather than a punishment to motivate effort, and with one exception that we describe momentarily. Consequently, the definition of first-order rotation must be amended: here rankings update after a bad output rather than a good output; we provide the details in Appendix B.3. There is one substantive difference. In this environment, improving effort detection unambiguously expands the scope of efficiency, because it raises the likelihood that effort produces a good output and slows rank transitions, thereby increasing the severity of the punishment associated with a bad output and strengthening incentives.

Time-varying tasks. We have interpreted our model as the manager being endowed with a single task and repeatedly assigning it to some workers. We could have equivalently interpreted the model as the manager repeatedly assigning a new but homogeneous task. Focusing on this latter interpretation, our results extend in a variant of our baseline model where the tasks are heterogeneous so that in each period, there are two possible task realizations, one undesirable and the other desirable.¹⁵ The analysis combines the logic in our undesirable-task baseline model and our desirable-task extension above; we provide the details in Appendix B.4.

Multiple tasks. Consider next a version of our baseline model where the manager has multiple tasks and must assign several workers to each perform a task in each period. For simplicity, suppose that there are two tasks and two workers are assigned in each period, and that both tasks are undesirable, with one being more undesirable than the other.¹⁶ The model is otherwise unchanged. Our results extend, with an additional feature that the evolution of assignment depends jointly on the outputs of both incumbent workers. Accordingly, the definition of first-order rotation is amended as follows. In each period, workers continue to be strictly ranked; the lowest ranked worker is assigned the most undesirable task, whereas the second lowest ranked worker is assigned the other, less undesirable task. If both incumbent workers produce good outputs, they are assigned the highest and the second highest ranks respectively with equal probability; if both produce bad outputs, they are assigned the two lowest ranks respectively with equal probability; if one produces good output and the other produces bad output, then the successful worker becomes top ranked while the unsuccessful

¹⁵Our results extend with more task realizations in the natural way.

¹⁶Our arguments extend to more tasks and to one or both of the tasks being desirable in the natural way.

worker is assigned the lowest rank. In all cases, the currently unassigned workers move down by the minimally necessary number of ranks. The uniform tie-breaking when both incumbent workers produce the same output arises because, as in the baseline model, workers must face identical incentives when moral hazard is maximally tight. We provide the details in Appendix B.5.

Limited worker information. In some applications, such as gig-economy platforms, workers may not observe who is assigned the task or the resulting output, except when they themselves are selected. Consider a variant of the baseline model in which, in each period, unassigned workers do not observe the identity of the assigned worker or his output; the model is otherwise unchanged. In this environment, it is natural to allow the manager and the workers to employ private strategies, rather than to examine perfect public equilibrium. We accordingly adopt belief-free equilibrium (Ely, Hörner and Olszewski, 2005) as the solution concept, requiring that players’ strategies remain optimal irrespective of their private histories. This solution concept is consistent with the uniformity criterion, as it rules out assignment rules that rely on the manager’s precise knowledge of workers’ beliefs or on the manager relentlessly tracking workers’ belief dynamics. Because belief-free equilibrium preserves a recursive structure, our results extend following analogous arguments in the baseline model.

Workers may also be uncertain about the true workforce size n . Accordingly, suppose further that the manager has private information about n , while workers hold a common prior belief on n . This setting features incomplete information and we extend the notion of belief-free equilibrium to Yamamoto’s (2014) ex post belief-free equilibrium, requiring that players’ strategies remain optimal irrespective of their private histories and the true state n . This solution concept likewise is consistent with the uniformity criterion and preserves a recursive structure, and our results continue to hold. Strictly speaking, both Ely et al. (2005) and Yamamoto (2014) defined belief-free equilibria for two-player finite games only, but their definitions can be adopted here in the natural way.

In settings captured by the preceding paragraphs, workers are uncertain about their relative ranks in each period under first-order rotation. This discussion therefore implies that the efficiency-equality tension in our model is robust to rank uncertainty.

8 Concluding comments

In this paper, we have examined task assignment among workers as an incentive instrument under dynamic moral hazard and also elucidated novel features of the resulting optimal incentives. While a pure moral hazard setting is a natural point of departure, in some applications workers may have private information about their ability that plays a nonnegligible role in the manager's assignment. To fix ideas, consider again our baseline, undesirable-task, model. Here, if each worker might be a type that is inherently unproductive in delivering a good output, then a manager who is sufficiently pessimistic about this worker's ability might choose to not assign him even after he produced a bad output; this undermines the scope of punishment. A systematic analysis of how the workers' private information shapes assignment dynamics requires distinct technical tools, as the recursive equilibrium structure that we have exploited in this paper does not apply and it is not satisfactory to adopt a belief-free solution concept that restores the recursive structure at the expense of shutting down the manager's reliance on her belief in determining assignments. We leave this to future research.

A Proofs

A.1 Proof of Lemma 1

Fix an efficient equilibrium and history h_t where worker i is assigned with positive probability. Worker i 's continuation payoff by exerting effort is given by

$$U^i(h_t i) = (1 - \delta) \times 0 + (pU^i(h_t i \bar{y}) + (1 - p)U^i(h_t i \underline{y})).$$

By deviating to shirk, his continuation payoff is

$$\hat{U}^i(h_t i) = (1 - \delta) \times s + (qU^i(h_t i \bar{y}) + (1 - q)U^i(h_t i \underline{y})).$$

By one-shot deviation principle, $U^i(h_t i) - \hat{U}^i(h_t i) \geq 0$. Equivalently, (3) holds.

Next, in each period in the continuation following history h_t , one worker is assigned and exerts effort while the other $n - 1$ workers are unassigned. Because in each such period the worker who is assigned and exerts effort gets payoff 0 while each unassigned worker gets payoff r , (4) follows.

A.2 Proof of Theorem 1

Fix $n > 1$. We proceed in three steps.

Step 1: Characterizing $\Omega^*(n)$

The first step is to prove that the set $\Omega^*(n)$ satisfies (7). We prove (7) via a series of lemmas. Define

$$\xi := \frac{1 - \delta(1 - p)}{\delta p}, \tag{9}$$

and then define

$$V^1(n, p, r, \delta) := \frac{\xi^{n-1} - 1}{\xi^n - 1} r, \quad (10)$$

$$V^k(n, p, r, \delta) := \left(1 - \frac{(\xi - 1)\xi^{k-2}}{\xi^n - 1}\right) r, \quad \text{for } k = 2, \dots, n. \quad (11)$$

When no risk of ambiguity arises, for each $k = 1, \dots, n$, we write $V^k(n, p, r, \delta)$ as V^k . We characterize $\text{bd}(\Omega^*(n))$. As is standard, for any $\omega \in \text{bd}(\Omega^*(n))$, it is without loss of generality to assume that the manager plays a pure strategy, and in the following we will do so. This is because E_ω^* is convex, as the manager's payoff is constant on E_ω^* and is willing to randomize over any efficient continuations, and because $\omega \in \text{bd}(\Omega^*(n))$ so that continuation payoff vectors emerging on path in any efficient equilibrium correspond to extreme points of E_ω^* , which in turn correspond to the manager playing pure strategies.

Recall that $\Omega^*(n)$ is closed and so contains its boundary. Consequently, an efficient equilibrium exists for any $\omega \in \text{bd}(\Omega^*(n))$.

Lemma 2. *Let $\omega \in \text{bd}(\Omega^*(n))$. In any efficient equilibrium, at any history h_t following which worker i is assigned,*

$$U^i(h_t) \geq \frac{ps}{p - q}. \quad (12)$$

Proof of Lemma 2. Fix $\omega \in \text{bd}(\Omega^*(n))$ and an efficient equilibrium. By Lemma 1, at any history h_t , the assigned worker i 's incentive constraint for effort holds:

$$U^i(h_t i \bar{y}) - U^i(h_t i \underline{y}) \geq \frac{(1 - \delta)s}{\delta(p - q)}. \quad (13)$$

Therefore, by definition of $\underline{U}$,

$$\begin{aligned} U^i(h_t) &= \delta(pU^i(h_t i \bar{y}) + (1 - p)U^i(h_t i \underline{y})) \\ &\geq \delta \left(U^i(h_t i \underline{y}) + \frac{(1 - \delta)ps}{\delta(p - q)} \right) \\ &\geq \delta \underline{U} + \frac{(1 - \delta)ps}{(p - q)}. \end{aligned}$$

Because the efficient equilibrium, the history h_t , and the assigned worker i are arbitrarily

picked, the above inequality holds when they are picked such that $U^i(h_t)$ attains $\underline{U}$, giving

$$\underline{U} \geq \delta \underline{U} + \frac{(1-\delta)ps}{(p-q)} \implies \underline{U} \geq \frac{ps}{(p-q)}. \quad (14)$$

Because $U^i(h_t) \geq \underline{U}$, (14) implies (12), as was to be shown. $\blacksquare$

Lemma 3. *Let $\omega \in \text{bd}(\Omega^*(n))$. It holds that*

$$\bar{U} - \underline{U} = \frac{(1-\delta)s}{\delta(p-q)}. \quad (15)$$

In any efficient equilibrium, for each $i = 1, \dots, n$, at each history where worker i is assigned, upon producing a good output, his continuation payoff is equal to $\bar{U}$; upon producing a bad output, his continuation payoff is $\underline{U}$.

Proof of Lemma 3. Fix $\omega \in \text{bd}(\Omega^*(n))$ and an efficient equilibrium. By Lemma 1, at history h , upon worker i being assigned, $U^i(hi\underline{y}) \geq \underline{U}$. Next, at any history h' , because there must be one worker who is assigned and then exerts effort, $U^i(h') \leq \bar{U}$. Consequently, $U^i(hi\bar{y}) \leq \bar{U}$. Therefore (3) implies

$$\bar{U} - \underline{U} \geq \frac{(1-\delta)s}{\delta(p-q)}. \quad (16)$$

To complete the proof, it suffices to show that this inequality cannot be strict. If it were strict, then by continuity in ω on both sides of (16), there exists an open ball B_ω containing ω such that an efficient assignment rule exists given each $\omega' \in B_\omega$, contradicting that $\omega \in \text{bd}(\Omega^*(n))$. $\blacksquare$

Lemma 4. *Let $\omega \in \text{bd}(\Omega^*(n))$. In any efficient equilibrium, for each $i = 1, \dots, n$, at each history h where worker i is assigned, his continuation payoff $U^i(hi)$ is equal to $\underline{U}$.*

Proof of Lemma 4. Fix $\omega \in \text{bd}(\Omega^*(n))$ and an efficient equilibrium, history h , and worker i assigned at this history. At this history, worker i 's continuation payoff $U^i(hi)$ upon being assigned is

$$(1-\delta) \times 0 + \delta(p\bar{U} + (1-p)\underline{U}). \quad (17)$$

This is because, by Lemma 3, $U^i(hi\bar{y}) = \bar{U}$ and $U^i(hi\underline{y}) = \underline{U}$. Therefore, (15) and (17)

together imply that

$$U^i(hi) = (1 - \delta) \frac{ps}{p - q} + \delta \underline{U}. \quad (18)$$

Because $U^i(h) \geq \underline{U}$ by definition of $\underline{U}$, (18) implies

$$(1 - \delta) \frac{ps}{p - q} + \delta \underline{U} \geq \underline{U} \implies \underline{U} \leq \frac{ps}{p - q}. \quad (19)$$

This and (14) together imply that $\underline{U} = ps/(p - q)$. Consequently, by (18),

$$U^i(h) = (1 - \delta) \frac{ps}{p - q} + \delta \frac{ps}{p - q} = \frac{ps}{p - q} = \underline{U},$$

as desired. ■

Lemma 5. Fix $\omega \in \text{bd}(\Omega^*(n))$. It holds that $\bar{U} = U^{(1)}$.

Proof. Fix $\omega \in \text{bd}(\Omega^*(n))$. In any efficient equilibrium, in each period, the assigned worker must have continuation payoff $\underline{U}$ by Lemma 4. Therefore, for any efficient payoff vector $U \in E_\omega^*$, at least one entry in U is equal to $\underline{U}$. Therefore

$$\begin{aligned} \bar{U} &= \max_{(U^1, \dots, U^n) \in E_\omega^*} U^1 \\ &= \max_{(U^1, \dots, \underline{U}) \in E_\omega^*} U^1 \\ &= U^{(1)}, \end{aligned}$$

as desired. ■

Lemma 6. Fix $\omega \in \text{bd}(\Omega^*(n))$. Then (5) holds. Moreover, in (5), $\underline{U} = V^1$ and $U^{(1)} = V^2$, given by (10) and (11).

Proof of Lemma 6. Let $\omega \in \text{bd}(\Omega^*(n))$ and $U^* = (\underline{U}, U^{(1)}, \dots, U^{(n-1)}) \in E_\omega^*$. Define $U' := (U^{(1)}, \dots, U^{(n-1)}, \underline{U})$. By symmetry, and because $U^* \in E_\omega^*$, $U' \in E_\omega^*$. Define also

$$\lambda(y) := \begin{cases} U', & \text{if } y = \bar{y}, \\ U^*, & \text{if } y = \underline{y}. \end{cases}$$

To prove the lemma, we must show that

$$U = (1 - \delta)(0, r, \dots, r) + \delta[p\lambda(\bar{y}) + (1 - p)\lambda(\underline{y})]. \quad (20)$$

Because $U, U' \in E_\omega^*$ and (15) holds, there is $\hat{U} := (\hat{U}_1, \dots, \hat{U}_n) \in E_\omega^*$ such that

$$\hat{U} = (1 - \delta)(0, r, \dots, r) + \delta[p\lambda(\bar{y}) + (1 - p)\lambda(\underline{y})]. \quad (21)$$

To complete the proof, it suffices to show that $\hat{U} = U^*$. By (21),

$$\hat{U}_1 = \delta(pU^{(1)} + (1 - p)U) = U, \quad (22)$$

where the second equality uses Lemma 3 and Lemma 4. For any output y , let $\lambda_i(y)$ denote the i th entry of $\lambda(y)$. Because $\lambda_1(\bar{y}) = U^{(1)}$ and $\lambda_1(\underline{y}) = U$, $\lambda_2(\bar{y}) \leq U^{(2)}$ and $\lambda_2(\underline{y}) \leq U^{(1)}$. Because each worker's per-period payoff in any efficient equilibrium is at most r , and because $U^* \in E_\omega^*$,

$$U^{(1)} \leq (1 - \delta)r + \delta(pU^{(2)} + (1 - p)U^{(1)}) = \hat{U}_2 \leq U^{(1)},$$

where the last inequality uses the definition of $U^{(1)}$. Therefore $\hat{U}_2 = U^{(1)}$. Then, by induction, it holds that for each $k = 1, \dots, n - 2$,

$$U^{(k)} = (1 - \delta)r + \delta[pU^{(k+1)} + (1 - p)U^{(k)}] = \hat{U}_{k+1} \leq U^{(k)},$$

and therefore $U^{(k)} = \hat{U}_{k+1}$. Finally, because $\lambda(\bar{y}), \lambda(\underline{y}) \in E_\omega^*$, it follows that

$$U^{(n-1)} = (1 - \delta)r + \delta[pU + (1 - p)U^{(n-1)}],$$

and therefore $U^{(n-1)} = \hat{U}_n$. Thus, $\hat{U} = U^*$, as desired. ■

Finally, it is straightforward to verify that for each $\omega \in \text{bd}(\Omega^*(n))$, $U^* = (V^1, \dots, V^n)$, given in (10) and (11), is the unique solution to (5). Therefore

$$\text{bd}(\Omega^*(n)) = \left\{ \omega \in \Omega(n) : V^2(n, p, r, \delta) - V^1(n, p, r, \delta) = \frac{(1 - \delta)s}{\delta(p - q)} \right\}, \quad (23)$$

which implies (7), as desired.

Step 2: First-order rotation rules are uniformly efficient

We next prove that all first-order rotation rules are uniformly efficient. Fix $\omega \in \Omega^*(n)$ and a strategy profile in which each worker exerts effort when assigned and the manager chooses some first-order rotation rule. This strategy profile is stationary and symmetric across workers. Recall that under this strategy profile, in each period on path, the continuation payoff of a worker with rank π is denoted by $U(\pi)$. The vector $(U(\pi_n), U(\pi_1), \dots, U(\pi_{n-1}))$ is therefore a unique solution to the system

$$\begin{pmatrix} U(\pi_n) \\ U(\pi_1) \\ \vdots \\ U(\pi_{n-1}) \end{pmatrix} = (1 - \delta) \begin{pmatrix} 0 \\ r \\ \vdots \\ r \end{pmatrix} + \delta \left[p \begin{pmatrix} U(\pi_1) \\ U(\pi_2) \\ \vdots \\ U(\pi_n) \end{pmatrix} + (1 - p) \begin{pmatrix} U(\pi_n) \\ U(\pi_1) \\ \vdots \\ U(\pi_{n-1}) \end{pmatrix} \right]. \quad (24)$$

Because $(V^1, \dots, V^n)$, given in (10) and (11), is the unique solution to (5), and because (5) coincides with (24), it follows that

$$(U(\pi_n), U(\pi_1), \dots, U(\pi_{n-1})) = (V^1, V^2, \dots, V^{n-1}, V^n).$$

Moreover,

$$U(\pi_1) - U(\pi_n) = V^2 - V^1 \geq \frac{(1 - \delta)s}{\delta(p - q)},$$

where the inequality uses the characterization of $\Omega^*(n)$ in Step 1. Consequently, the strategy profile constitutes an equilibrium. Because the fixed first-order rotation rule is arbitrarily chosen, given $\omega \in \Omega^*(n)$, first-order rotation rules are efficient. Because first-order rotation rules do not depend on ω , the above arguments hold uniformly over all $\omega \in \Omega^*(n)$. Therefore first-order rotation rules are uniformly efficient, as desired.

Step 3: Uniform efficiency implies first-order rotation

Finally, we prove that all uniformly efficient assignment rules must prescribe first-order rotation. Fix any $\omega \in \text{bd}(\Omega^*(n))$ and any efficient equilibrium. The arguments in the first step imply that, in each period on path, the assigned worker's, say worker 1's, continuation payoff is $\underline{U}$; a good output raises this payoff to $U^{(1)}$, whereas a bad output leaves it unchanged. Once worker 1's continuation payoff reaches $U^{(1)}$, he is unassigned and some other worker, say worker 2, is assigned with continuation payoff $\underline{U}$. If there are only two workers, then a good output by worker 2 transitions their payoffs from $(U^{(1)}, \underline{U})$ to $(\underline{U}, U^{(1)})$, while a bad output leaves them unchanged. The transition for worker 2's payoff follows the logic described above, whereas the transition for worker 1's payoff follows because, given that worker 2 is granted respite, worker 1 must be assigned. If there are more than two workers, then a good output by worker 2 transitions their payoffs from $(U^{(1)}, \underline{U})$ to $(U^{(2)}, U^{(1)})$, while a bad output leaves them unchanged. The transition for worker 2's payoff follows the logic described above, whereas the transition for worker 1's payoff holds because, given worker 2's payoff transition, any alternative payoff transition for worker 1 would make him worse off, contradicting that $U^{(1)}$ is the highest continuation payoff a worker can obtain in any efficient equilibrium. When these workers hold continuation payoffs $(U^{(2)}, U^{(1)})$, they are unassigned and a third worker must be assigned with continuation payoff $\underline{U}$. Iterating this argument shows that for each worker, conditional on currently being assigned, the continuation payoff starts at $\underline{U}$, moves cyclically through $U^{(1)}, \dots, U^{(n-1)}$ before returning to $\underline{U}$, changing value after a good output but staying put after a bad output. Consequently, in any efficient equilibrium, a worker who is currently assigned and then produces a good output rests until each other worker has also produced a good output. The manager's strategy must therefore prescribe first-order rotation.

The three steps together imply Theorem 1, completing the proof.

A.3 Proof of Proposition 1

From the proof of Theorem 1, $U(\pi_k) = V^{k+1}$ and $U(\pi_n) = V^1$. By (10) and (11), it follows from direct computation that $V^2 > V^3 > \dots > V^n > V^1$, leading to Proposition 1.

A.4 Proof of Proposition 2

From the proof of Theorem 1, $U(\pi_k) = V^{k+1}$ and $U(\pi_n) = V^1$. Recall from Equation (9) that

$$\xi = \frac{1 - \delta(1 - p)}{\delta p} > 1.$$

In this proof, we write ξ as $\xi(p)$ to emphasize its dependence on p . By (10) and (11),

$$S(n, p, q, r) = r \frac{(\xi(p) - 1)(\xi(p)^{n-1} - 1)}{(\xi(p)^n - 1)} - \frac{(1 - \delta)s}{\delta(p - q)}. \quad (25)$$

Part 1 is immediate. Here we prove part 2. Write $S(n, p, q, r)$, given in (25), as $F(p)$ to stress its dependence on p . We show that F admits a unique maximizer on $(q, 1]$. Let $a := 1 - \delta \in (0, 1)$. Define $A(p) := a + \delta p$, $x(p) := \delta p / A(p)$, $Q(z) := (1 - z^{n-1}) / (1 - z^n)$, and $\kappa := s / (r\delta)$. By direct calculations,

$$F(p) \propto r \cdot \frac{Q(x(p))}{A(p)} - \frac{1}{\delta} \cdot \frac{s}{p - q}.$$

Maximizing F is equivalent to maximizing

$$\phi(p) := \frac{Q(x(p))}{A(p)} - \frac{\kappa}{p - q}.$$

Because $\phi(p) \rightarrow -\infty$ as $p \rightarrow q$ and ϕ is continuous on $(q, 1]$, a maximizer exists on $(q, 1]$. We prove it is unique. Differentiating ϕ yields $\phi'(p) = R(p) - L(p)$, where on $(q, 1]$,

$$R(p) := \kappa \cdot \frac{1}{(p - q)^2}, \quad L(p) := -\left(\frac{Q(x(p))}{A(p)}\right)'.$$

Note that R is strictly decreasing. Moreover, $\lim_{p \searrow q} R(p) = +\infty$, and so $\lim_{p \searrow q} \phi'(p) = +\infty$. Thus, if L is strictly increasing, the equation $\phi'(p) = 0$ has at most one solution. Consequently, the maximizer of ϕ is unique.

It remains to show that L is strictly increasing. Since $x(p)$ is strictly increasing in p , it is a bijection from $(0, 1]$ onto $(0, \bar{x}]$ where $\bar{x} := x(1) = \delta \in (0, 1)$, and monotonicity in p is equivalent to monotonicity in x . Recall that $x(p) = \delta p / (a + \delta p)$. Solving $x = \delta p / (a + \delta p)$ for

p yields

$$p = \frac{a}{\delta} \cdot \frac{x}{1-x}, \quad \text{and} \quad A(p) = a + \delta p = \frac{a}{1-x}.$$

Therefore $Q(x)/A(p) = (1-x)Q(x)/a$. Let $W(x) := (1-x)Q(x)$. Then, it holds that $L(p(x)) = -\delta(1-x)^2 W'(x)/a$. Thus it suffices to show that the function $\Lambda(x) := -(1-x)^2 W'(x)$ is strictly increasing on $(0, \bar{x}]$.

Write $G_m(x) := \sum_{k=0}^m x^k$. Then $Q(x) = G_{n-2}(x)/G_{n-1}(x)$ and

$$W(x) = (1-x) \frac{G_{n-2}(x)}{G_{n-1}(x)}.$$

By direct calculation, and using $G_{n-1} = G_{n-2} + x^{n-1}$, it follows that

$$Q'(x) = \frac{G'_{n-2}(x)G_{n-1}(x) - G_{n-2}(x)G'_{n-1}(x)}{G_{n-1}(x)^2} = -\frac{x^{n-2}}{G_{n-1}(x)^2} < 0. \quad (26)$$

Differentiating W , and using (26), yields

$$-W'(x) = \frac{G_{n-2}(x)}{G_{n-1}(x)} + \frac{(1-x)x^{n-2}}{G_{n-1}(x)^2} = \frac{G_{n-2}(x)G_{n-1}(x) + (1-x)x^{n-2}}{G_{n-1}(x)^2}.$$

Hence

$$\Lambda(x) = (1-x)^2 \frac{G_{n-2}(x)G_{n-1}(x) + (1-x)x^{n-2}}{G_{n-1}(x)^2}. \quad (27)$$

Differentiating Λ then yields

$$\Lambda'(x) = (1-x) \frac{P_n(x)}{G_{n-1}(x)^3},$$

where $P_n(x)$ is a polynomial with nonnegative coefficients and at least one strictly positive coefficient. Consequently $\Lambda'(x) > 0$ for all $x \in (0, 1)$, as was to be shown.

To complete the proof, we show that there exists $\underline{r} > 0$ such that this maximizer is interior if $r > \underline{r}$ and is equal to 1 otherwise. Let $J := \sum_{k=0}^{n-1} \delta^k$, $T := \sum_{k=0}^{n-2} \delta^k$, and let p^* be the unique maximizer of F on $(q, 1]$. We show that $p^* \in (q, 1)$ if

$$s < r(1-q)^2 \left(\delta^2 \frac{T}{J} + (1-\delta) \frac{\delta^n}{J^2} \right). \quad (28)$$

and $p^* = 1$ otherwise. Since $\phi'(p) \rightarrow +\infty$ as $p \downarrow q$, the unique maximizer is interior if $\phi'(1) < 0$, and is equal to 1 otherwise.

Recall that $W(p) = Q(x(p))/A(p)$. Then $\phi'(p) = W'(p) + \kappa/(p - q)^2$, so $\phi'(1) < 0$ is equivalent to

$$\frac{\kappa}{(1 - q)^2} < -W'(1). \quad (29)$$

We compute $W'(1)$. Differentiating $W(p) = Q(x(p))/A(p)$ gives

$$W'(p) = \frac{Q'(x(p))x'(p)}{A(p)} - \frac{Q(x(p))A'(p)}{A(p)^2},$$

and so $W'(1) = Q'(\delta)\delta(1 - \delta) - Q(\delta)\delta$. Recall $Q(x) = G_{n-2}(x)/G_{n-1}(x)$ and $Q'(x) = -x^{n-2}/G_{n-1}(x)^2$. With $J = G_{n-1}(\delta)$ and $T = G_{n-2}(\delta)$, this yields

$$-W'(1) = \delta \frac{T}{J} + (1 - \delta) \frac{\delta^{n-1}}{J^2}.$$

Substituting this into (29), recalling $\kappa = s/(r\delta)$, and rearranging, (28) follows. Finally, if (28) fails, or equivalently $\phi'(1) \geq 0$, then single-peakedness of ϕ implies $p^* = 1$, as was to be shown.

A.5 Proofs of Proposition 3

In this proof, we again write ξ as $\xi(p)$ to emphasize its dependence on p . By (10) and (11),

$$I(k, k'; n, p, r) = \begin{cases} r \frac{(\xi(p) - 1)(\xi(p)^{n-1} - \xi(p)^{k-2})}{(\xi(p)^n - 1)}, & \text{if } k' = n \text{ and } k < n, \\ r \frac{(\xi(p) - 1)(\xi(p)^{k'-2} - \xi(p)^{k-2})}{(\xi(p)^n - 1)}, & \text{if } n > k' > k, \end{cases} \quad (30)$$

where in $I(k, k'; n, p, r)$ we have ignored the case $k = n$ and also the case $k > k'$ because it is symmetric in k and k' . Parts 1—3 are immediate given (30). We prove part 4 below. Fix a distinct pair (k, k') . Without loss of generality, let $k' > k$. Write (30) as $I(\xi)$ in this proof to emphasize its dependence on ξ . We consider two cases in order.

1. Let $k' = n$. Then

$$I(\xi) \propto \frac{(\xi - 1)(\xi^{n-1} - \xi^{k-2})}{\xi^n - 1}.$$

By direct calculations, $I'(\xi) > 0$ if and only if $J(\xi) := \xi^{n-k+2} + (n-k)(\xi - 1) - 1 > 0$. This latter inequality holds because $J' > 0$ and $J(1) = 0$. Because $\xi'(p) < 0$ by (9), part 1 follows.

2. Let $k' < n$, $k < k'$, and $a := k' - k \in \{1, \dots, n-2\}$. Because $I(\xi) > 0$, the sign of $I'(\xi)$ equals the sign of the log-derivative of I . Let $\phi(\xi) := \log I(\xi)$. By direct calculations,

$$\phi'(\xi) = \frac{1}{\xi - 1} + \frac{k-2}{\xi} + \frac{a\xi^{a-1}}{\xi^a - 1} - \frac{n\xi^{n-1}}{\xi^n - 1}. \quad (31)$$

For integers $m = 1, 2, \dots$, define

$$g_m(\xi) := \frac{m\xi^{m-1}}{\xi^m - 1} = \frac{m}{\xi} \cdot \frac{1}{1 - \xi^{-m}} = \frac{m}{\xi} \sum_{\ell=0}^{\infty} \xi^{-m\ell}.$$

Then from (31),

$$\phi'(\xi) = \frac{1}{\xi - 1} + \frac{k-2}{\xi} + g_a(\xi) - g_n(\xi). \quad (32)$$

Because $\xi'(p) < 0$ by (9), to complete the proof, it suffices to show that $\phi'' < 0$, $\lim_{\xi \searrow 1} \phi'(\xi) = +\infty$, and $\phi'(\xi) \sim \hat{\phi}$ for some $\hat{\phi} < 0$ as $\xi \rightarrow \infty$, where $\sim$ denotes asymptotic equivalence. By (32),

$$\begin{aligned} \phi''(\xi) &= -\frac{1}{(\xi - 1)^2} - \frac{k-2}{\xi^2} + (g_a - g_n)'(\xi) \\ &< (g_a - g_n)'(\xi) \\ &= \sum_{\ell=0}^{\infty} [-a(a\ell + 1)\xi^{-(a\ell+2)} + n(n\ell + 1)\xi^{-(n\ell+2)}] \\ &< 0. \end{aligned}$$

Next, it is immediate to see from (31) that $\lim_{\xi \searrow 1} \phi'(\xi) = +\infty$. Finally, as $\xi \rightarrow \infty$,

$g_m(\xi) \sim m/\xi$ and $1/(\xi - 1) \sim 1/\xi$. Hence, by (31),

$$\phi'(\xi) \sim \frac{1}{\xi} + \frac{k-2}{\xi} + \frac{a}{\xi} - \frac{n}{\xi} = \frac{k+a-n-1}{\xi} = \frac{k'-n-1}{\xi} < 0,$$

as desired.

B Omitted details

B.1 Example of probabilistic reward

In this Appendix, we provide an example of an assignment rule that serves two purposes. First, it illustrates an assignment rule that is efficient for a range of parameter vectors but not uniformly efficient. Second, it elaborates on the use of probabilistic reward discussed in Section 4.

Suppose that there are two workers. Consider an assignment rule where worker 1 is initially assigned. In each period, if the assigned worker delivers a good output, then in the next period the manager assigns the other worker with some probability ξ , chosen below to be such that the currently assigned worker is indifferent between exerting effort and shirking in the corresponding efficient equilibrium, and retains the currently assigned worker otherwise. If the assigned worker delivers a bad output, then he is retained. Fix a strategy profile where the manager plays this assignment rule and the workers exert effort when assigned. Write $\hat{U}_S$ as the assigned worker's continuation payoff and $\hat{U}_R$ as the unassigned worker's continuation payoff in each period under this strategy profile. The payoffs $(\hat{U}_S, \hat{U}_R)$ and the transition probability ξ jointly solve the system of promise-keeping constraints

$$\begin{aligned} \xi(\hat{U}_R - \hat{U}_S) &= \frac{(1-\delta)s}{\delta(p-q)}, \\ \hat{U}_S &= \delta(p\xi\hat{U}_R + (1-p\xi)\hat{U}_S), \\ \hat{U}_R &= (1-\delta)r + \delta(p\xi\hat{U}_S + (1-p\xi)\hat{U}_R). \end{aligned}$$

The first equation is the assigned worker's incentive constraint. It is binding because the transition probability upon a good output is chosen to be sufficiently low so that the expected reward from a good output is just enough to make the worker indifferent between exerting

effort and shirking whenever assigned. The second and third equations are the workers' promise-keeping constraints. It can be readily verified that when the shirking gain s is sufficiently small, there is a solution to this system, so that the assignment rule constitutes an efficient equilibrium. Nonetheless, in this equilibrium, the transition probability ξ lies in $(0, 1)$ and depends nontrivially on the parameter vector ω for each ω on the interior of $\Omega^*(n)$. Consequently, this assignment rule, while efficient, is not uniformly efficient.

B.2 Effort detection on efficiency and inequality

In this section, we provide a sufficient condition on the parameters under which a marginal improvement in effort detection.

Proposition 4. *There exists $\underline{\delta} \in (0, 1)$ such that for each $\delta \geq \underline{\delta}$, the scope of efficiency $S(n, p, q, r)$ is strictly increasing in p and the inequality $I(\cdot; n, p, r)$ is strictly decreasing in p .*

Proof of Proposition 4. Define

$$f_S(\delta) := \frac{\partial S(n, p, r)}{\partial p}.$$

By direct calculations,

$$\begin{aligned} \lim_{\delta \rightarrow 1} f_S(\delta) &= 0, \\ \lim_{\delta \rightarrow 1} f'_S(\delta) &= \frac{pns - (p - q)(n - 1)r}{np(p - q)}. \end{aligned}$$

There exists $\bar{p} \in (0, 1)$ such that for every $p \in (q, \bar{p})$, $\lim_{\delta \rightarrow 1} f'_S(\delta) < 0$. This shows that given sufficient high p , for δ sufficiently close to 1, $f_S(\delta) > 0$, as was to be shown. Next, for distinct k, k' , define

$$f_I(\delta) := \frac{\partial I(k, k'; n, p, r)}{\partial p}.$$

By direct calculations,

$$\begin{aligned} \lim_{\delta \rightarrow 1} f_I(\delta) &= 0, \\ \lim_{\delta \rightarrow 1} f'_I(\delta) &= \frac{(k' - k)r}{np^2} > 0. \end{aligned}$$

This shows that for δ sufficiently close to 1, $f_I(\delta) < 0$, as was to be shown. ■

B.3 Desirable task

In this section, we provide the omitted details regarding the desirable-task extension in Section 7. Our results extend following a modification of the definition of first-order rotation.

Definition 3 (First-order rotation with desirable task). *In this desirable-task setting, an assignment rule is a first-order rotation rule if it satisfies the following. In each period, workers are labeled $\pi_1, \dots, \pi_n$ and the worker currently labeled π_n is assigned. Upon a good output, all workers retain their current labels. Upon a bad output, worker currently labeled π_k is relabeled $\pi_{(k \bmod n)+1}$ for each $k = 1, \dots, n$.*

Extending Theorem 1 also requires minimal modifications in the proof. Here we point out the modifications. In the first step where we characterize the largest set of efficient parameter vectors $\Omega^*(n)$ given workforce size n , here, unlike in the baseline model, we directly define $U^{(1)}$ as the highest continuation payoff that a worker can obtain in any efficient equilibrium: $U^{(1)} := \max_{(U^1, \dots, U^n) \in E^*} U^1$. Then, iterating for $k = 2, \dots, n$, we define the highest continuation payoff a worker can obtain in any efficient equilibrium conditional on there being k other workers each with continuation payoff $U^{(1)}, \dots, U^{(k-1)}$ by $U^{(k)} := \max_{(U^{(1)}, \dots, U^{(k-1)}, U^k, \dots) \in E^*} U^k$. By construction, both $(U^{(1)}, U^{(2)}, \dots, U^{(n)})$ and $(U^{(n)}, U^{(1)}, \dots, U^{(n-1)})$ are efficient equilibrium payoff vectors. For the same reason as in (5), for each $\omega \in \text{bd}(\Omega^*(n))$, these vectors satisfy

$$\begin{pmatrix} U^{(1)} \\ U^{(2)} \\ \vdots \\ U^{(n)} \end{pmatrix} = (1 - \delta) \begin{pmatrix} 0 \\ r \\ \vdots \\ r \end{pmatrix} + \delta \left[p \begin{pmatrix} U^{(1)} \\ U^{(2)} \\ \vdots \\ U^{(n)} \end{pmatrix} + (1 - p) \begin{pmatrix} U^{(n)} \\ U^{(1)} \\ \vdots \\ U^{(n-1)} \end{pmatrix} \right]. \quad (33)$$

Writing $(\hat{V}^k(n, p, r, \delta))_{k=1}^n$ as the value of $(U^{(k)})_{k=1}^n$ that solves the system (33), the same arguments in the proof of Theorem 1 in the baseline model imply

$$\Omega^*(n) = \left\{ \omega \in \Omega(n) : \hat{V}^2(n, p, r, \delta) - \hat{V}^1(n, p, r, \delta) \geq \frac{(1 - \delta)s}{\delta(p - q)} \right\}.$$

The second and last steps are virtually identical.

B.4 Time-varying tasks

In this section, we provide the omitted details regarding the time-varying-task extension in Section 7. Specifically, the model is identical to our baseline model except that in each period, unassigned workers receive the same payoff $r_+ > 0$ with probability $\gamma \in (0, 1)$ and $r_- < 0$ with complementary probability. Consequently, the former event captures that the current task is undesirable whereas the latter captures that it is desirable. The expected payoff of being unassigned is therefore $\gamma r_+ + (1 - \gamma)r_-$. To facilitate comparison with our baseline model and the above desirable-task extension, we abuse notation, denoting this expected payoff as r if it is positive and as $-r$ if it is negative. As in our baseline model, higher r continues to represent stronger worker preference between being assigned and being unassigned.

As mentioned in the main text, our results extend following a modification of the definition of first-order rotation.

Definition 4 (First-order rotation with time-varying tasks). *In this heterogeneous-task setting, an assignment rule is a first-order rotation rule if it satisfies the following. In each period, workers are labeled $\pi_1, \dots, \pi_n$ and the worker currently labeled π_n is assigned. If the task is undesirable, then upon a bad output, all workers retain their current labels; upon a good output, worker currently labeled π_k is assigned label $\pi_{(k \bmod n)+1}$ for each $k = 1, \dots, n$. If the task is desirable instead, then upon a good output, all workers retain their current labels; upon a bad output, worker currently labeled π_k is relabeled $\pi_{(k \bmod n)+1}$ for each $k = 1, \dots, n$.*

Extending Theorem 1 also requires minimal modifications in the proof. Here we point out the modifications. In the first step where we characterize $\Omega^*(n)$, we define, like in Appendix B.3 but unlike in the baseline model, $U^{(1)}$ directly as the highest continuation payoff that a worker can obtain in any efficient equilibrium:

$$U^{(1)} := \max_{(U^1, \dots, U^n) \in E^*} U^1.$$

Then, iterating for $k = 2, \dots, n$, define the highest continuation payoff a worker can obtain in any efficient equilibrium conditional on there being k other workers each with continuation

payoff $U^{(1)}, \dots, U^{(k-1)}$ by

$$U^{(k)} := \max_{(U^{(1)}, \dots, U^{(k-1)}, U^k, \dots) \in E^*} U^k.$$

By construction, both $(U^{(1)}, U^{(2)}, \dots, U^{(n)})$ and $(U^{(n)}, U^{(1)}, \dots, U^{(n-1)})$ are efficient equilibrium payoff vectors. Further for each $k = 1, \dots, n$, $U^{(k)}$ is an expected continuation payoff, with the expectation taken over the current task desirability, so that it can be written as

$$U^{(k)} = \gamma U_+^{(k)} + (1 - \gamma) U_-^{(k)}, \quad (34)$$

with $U_+^{(k)}$ being the continuation payoff conditional on the current task being undesirable and $U_-^{(k)}$ being the counterpart conditional on the current task being desirable. Then, for each $\omega \in \text{bd}(\Omega^*(n))$, as in (5), given a realization of an undesirable task,

$$\begin{pmatrix} U_+^{(n)} \\ U_+^{(1)} \\ \vdots \\ U_+^{(n-1)} \end{pmatrix} = (1 - \delta) \begin{pmatrix} 0 \\ r_+ \\ \vdots \\ r_+ \end{pmatrix} + \delta \left[p \begin{pmatrix} U^{(1)} \\ U^{(2)} \\ \vdots \\ U^{(n)} \end{pmatrix} + (1 - p) \begin{pmatrix} U^{(n)} \\ U^{(1)} \\ \vdots \\ U^{(n-1)} \end{pmatrix} \right]. \quad (35)$$

Given a realization of a desirable task,

$$\begin{pmatrix} U_-^{(1)} \\ U_-^{(2)} \\ \vdots \\ U_-^{(n)} \end{pmatrix} = (1 - \delta) \begin{pmatrix} 0 \\ r_- \\ \vdots \\ r_- \end{pmatrix} + \delta \left[p \begin{pmatrix} U^{(1)} \\ U^{(2)} \\ \vdots \\ U^{(n)} \end{pmatrix} + (1 - p) \begin{pmatrix} U^{(n)} \\ U^{(1)} \\ \vdots \\ U^{(n-1)} \end{pmatrix} \right]. \quad (36)$$

Writing $(\hat{V}^k)_{k=1}^n$ as the value of $(U^{(k)})_{k=1}^n$ that solves (34)–(36), the same arguments in the proof of Theorem 1 in the baseline model imply

$$\Omega^*(n) = \left\{ \omega \in \Omega(n) : \hat{V}^2(n, p, r, \delta) - \hat{V}^1(n, p, r, \delta) - \frac{(1 - \delta)s}{\delta(p - q)} \geq 0 \right\}. \quad (37)$$

The second and last steps are virtually identical.

B.5 Multiple tasks

In this section, we provide the omitted details regarding the multiple-task extension in Section 7. Specifically, suppose that in each period, the unassigned worker gets payoff $r > 0$. The worker assigned for undesirable task 1 gets payoff 0 if he exerts effort and s if he shirks, as in the baseline model. The worker assigned for undesirable task 2 gets payoff $x \in (0, r)$ if he exerts effort and $x + s$ if he shirks, capturing that task 2 is less undesirable than task 1, and that workers prefer rest than being assigned and exerting effort on either task.

Our results extend following a modification of the definition of first-order rotation.

Definition 5 (First-order rotation with two undesirable tasks). *In this two-undesirable-task setting, an assignment rule is a first-order rotation rule if it satisfies the following. In each period, workers are labeled $\pi_1, \dots, \pi_n$ and the workers currently labeled π_{n-1} and π_n are assigned. If both produce bad outputs, then workers π_{n-1} and π_n swap their labels with probability $1/2$ and retain their labels otherwise; all other workers retain their current labels. If both produce good outputs, then worker currently labeled π_k for each $k = 1, \dots, n-2$ is assigned label π_{k+2} whereas workers currently labeled π_{n-1} and π_n are each relabeled, with equal probabilities, π_1 and π_2 .*

The incentive constraint (3) in Lemma 1 must also be modified because two outputs are produced in each period. In any equilibrium, at each history h_t where workers i and j are assigned, the incentive constraint for worker i is now

$$\begin{aligned} & p(U^i(h_t(i, j)(\bar{y}, \bar{y})) - U^i(h_t(i, j)(\underline{y}, \bar{y}))) \\ & + (1 - p)(U^i(h_t(i, j)(\bar{y}, \underline{y})) - U^i(h_t(i, j)(\underline{y}, \underline{y}))) \geq \frac{(1 - \delta)s}{\delta(p - q)}, \end{aligned} \quad (38)$$

where $U^i(h_t(i, j)(y_t^i, y_t^j))$ denotes worker i 's continuation payoff following output y_t^i from worker i and y_t^j from worker j at that history in the equilibrium.

Extending Theorem 1 requires minimal modifications in the proof. Here we point out the modifications. In the first step where we characterize the largest set of efficient parameter vectors $\Omega^*(n)$ given workforce size n , we define the lowest continuation payoff that a worker can obtain in any efficient equilibrium conditional on some other worker obtaining continuation payoff $\underline{U}$ by $\underline{U}' := \min_{(U^1, \dots, \underline{U}) \in E^*} U^1$. Next, define the highest continuation payoff that a worker can obtain in any efficient equilibrium conditional on some other workers obtaining

continuation payoffs $\underline{U}$ and $\underline{U}'$ by $U^{(1)} := \max_{(\underline{U}, \underline{U}', U^3, \dots) \in E^*} U^3$. Finally, iterating for $k = 2, \dots, n-2$, define the highest continuation payoff a worker can obtain in any efficient equilibrium conditional on there being $k+1$ other workers each achieving continuation payoffs $\underline{U}, \underline{U}', U^{(1)}, \dots, U^{(k-1)}$ by $U^{(k)} := \max_{(\underline{U}, \underline{U}', U^{(1)}, \dots, U^{(k-1)}, U^{k+2}, \dots) \in E^*} U^{k+2}$. By construction, any payoff vector in which the entries are permutations of $(\underline{U}, \underline{U}', U^{(1)}, \dots, U^{(n-2)})$ lies in E^* . To ease the exposition, we first suppose that there are at least four workers, before explaining at the end how to accommodate the case of three workers. For any parameter vector $\omega \in \text{bd}(\Omega^*(n))$, in each period where workers i and j are assigned, both of their incentive constraints (38) bind and therefore must be identical when being rewarded to the best extent possible upon a good output and being punished as harshly as possible upon a bad output. Consequently, when both workers produce good outputs, each of them gets expected payoff $\frac{1}{2}U^{(1)} + \frac{1}{2}U^{(2)}$, reflecting that the continuation payoff vector upon their good outputs transition with equal probabilities to payoff vectors where one of them gets $U^{(1)}$ and the other gets $U^{(2)}$. Similarly, when both workers produce a bad output, each of them gets expected payoff $\frac{1}{2}\underline{U} + \frac{1}{2}\underline{U}'$. Finally, when one worker produces a good output while the other produces a bad output, then the successful worker gets payoff $U^{(1)}$ while the unsuccessful worker gets payoff $\underline{U}$. Then, from the same arguments as in the proof of Theorem 1, the payoff vector $(\underline{U}, \underline{U}', U^{(1)}, \dots, U^{(n-2)})$ satisfies

$$\begin{pmatrix} \underline{U} \\ \underline{U}' \\ U^{(1)} \\ \vdots \\ U^{(n-2)} \end{pmatrix} = (1-\delta) \begin{pmatrix} 0 \\ x \\ r \\ \vdots \\ r \end{pmatrix} + \delta \left[p^2 \begin{pmatrix} \frac{1}{2}U^{(1)} + \frac{1}{2}U^{(2)} \\ \frac{1}{2}U^{(1)} + \frac{1}{2}U^{(2)} \\ U^{(3)} \\ \vdots \\ \underline{U} \end{pmatrix} + p(1-p) \begin{pmatrix} \underline{U} \\ U^{(1)} \\ U^{(2)} \\ \vdots \\ \underline{U}' \end{pmatrix} \right. \\ \left. + (1-p)p \begin{pmatrix} U^{(1)} \\ \underline{U} \\ U^{(2)} \\ \vdots \\ \underline{U}' \end{pmatrix} + (1-p)^2 \begin{pmatrix} \frac{1}{2}\underline{U} + \frac{1}{2}\underline{U}' \\ \frac{1}{2}\underline{U} + \frac{1}{2}\underline{U}' \\ U^{(1)} \\ \vdots \\ U^{(n-2)} \end{pmatrix} \right].$$

Writing $(\hat{V}^k(n, p, r, \delta))_{k=1}^n$ as the value of $(\underline{U}, \underline{U}', U^{(1)}, \dots, U^{(n-2)})$ that solves this system, the

same arguments in the proof of Theorem 1 in the baseline model imply

$$\Omega^*(n) = \left\{ \omega \in \Omega(n) : p \left(\frac{1}{2} \hat{V}^3(n, p, r, \delta) + \frac{1}{2} \hat{V}^4(n, p, r, \delta) - \hat{V}^1(n, p, r, \delta) \right) \right. \\ \left. + (1-p) \left(\hat{V}^3(n, p, r, \delta) - \frac{1}{2} \hat{V}^1(n, p, r, \delta) - \frac{1}{2} \hat{V}^2(n, p, r, \delta) \right) \geq \frac{(1-\delta)s}{\delta(p-q)} \right\}.$$

If there are only three workers, then the payoff vector $(\underline{U}, \underline{U}', U^{(1)})$ satisfies

$$\begin{pmatrix} \underline{U} \\ \underline{U}' \\ U^{(1)} \end{pmatrix} = (1-\delta) \begin{pmatrix} 0 \\ x \\ r \end{pmatrix} + \delta \left[p^2 \begin{pmatrix} \frac{1}{2} U^{(1)} + \frac{1}{2} \underline{U}' \\ \frac{1}{2} U^{(1)} + \frac{1}{2} \underline{U}' \\ \underline{U} \end{pmatrix} + p(1-p) \begin{pmatrix} \underline{U} \\ U^{(1)} \\ \underline{U}' \end{pmatrix} \right. \\ \left. + (1-p)p \begin{pmatrix} U^{(1)} \\ \underline{U} \\ \underline{U}' \end{pmatrix} + (1-p)^2 \begin{pmatrix} \frac{1}{2} \underline{U} + \frac{1}{2} \underline{U}' \\ \frac{1}{2} \underline{U} + \frac{1}{2} \underline{U}' \\ U^{(1)} \end{pmatrix} \right].$$

Again writing $(\hat{V}^k(n, p, r, \delta))_{k=1}^3$ as the value of $(\underline{U}, \underline{U}', U^{(1)})$ that solves this system, the same arguments in the proof of Theorem 1 imply

$$\Omega^*(n) = \left\{ \omega \in \Omega(n) : p \left(\frac{1}{2} \hat{V}^3(n, p, r, \delta) + \frac{1}{2} \hat{V}^2(n, p, r, \delta) - \hat{V}^1(n, p, r, \delta) \right) \right. \\ \left. + (1-p) \left(\hat{V}^3(n, p, r, \delta) - \frac{1}{2} \hat{V}^1(n, p, r, \delta) - \frac{1}{2} \hat{V}^2(n, p, r, \delta) \right) \geq \frac{(1-\delta)s}{\delta(p-q)} \right\}.$$

The second and last steps are virtually identical.

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